



Lab Assignment – 3 (SQL Querying & Logical Thinking)

Department: CSE

Subject: DIS Lab
Code: 22CPP217

Objective: The objective of this lab is to strengthen **SQL querying and analytical thinking** by applying filtering, sorting, grouping, and aggregation techniques. Students will translate real-world business requirements into **logical and optimized SQL queries** for data-driven decision making.

Reference : Aggregate Functions

Pre-Lab Setup : Run this block to clean your environment, disable "Safe Update" restrictions, and create the necessary tables.

SQL

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-- 0. Disable Safe Update Mode (Prevents errors on bulk updates)
SET SQL_SAFE_UPDATES = 0;

-- 1. Reset Tables
DROP TABLE IF EXISTS table_name;

-- 2. Create Main Table <student_registry> with following column names
<ID, Name, Dept_Name, Total_Credits, CGPA, Email>
-- Apply constraints-
    ID:primary key
    Name:not null

-- 3. Insert Data in the table from given SQL query below

INSERT INTO Student_Registry (ID, Name, Dept_Name, Total_Credits, CGPA,
Email) VALUES
(101, 'Aarav Sharma', 'Comp. Sci.', 85, 8.9, '101@univ.edu'),
(102, 'Priya Verma', 'Comp. Sci.', 72, 8.2, '102@univ.edu'),
(103, 'Rohit Singh', 'Comp. Sci.', 45, 6.8, NULL),
(104, 'Neha Gupta', 'Physics', 90, 9.1, '104@univ.edu'),
(105, 'Kunal Mehta', 'Physics', 30, 5.9, NULL),
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(106, 'Ananya Iyer', 'Chemistry', 78, 8.0, '106@univ.edu'),
(107, 'Siddharth Jain', 'Chemistry', 60, 7.4, NULL),
(108, 'Pooja Nair', 'Mathematics', 88, 9.3, '108@univ.edu'),
(109, 'Rahul Khanna', 'Mathematics', 40, 6.2, NULL),
(110, 'Ishita Roy', 'History', 35, 7.1, '110@univ.edu'),
(111, 'Vikram Patel', 'History', 20, 5.8, NULL),
(112, 'Sneha Kulkarni', 'Economics', 92, 8.6, '112@univ.edu'),
(113, 'Arjun Malhotra', 'Economics', 55, 7.9, NULL),
(114, 'Meera Joshi', 'General', 15, 6.5, NULL),
(115, 'Nitin Rao', 'General', 8, 5.2, NULL),
(116, 'Kavya Reddy', 'Comp. Sci.', 28, 6.0, NULL),
(117, 'Aditya Bose', 'Physics', 65, 7.8, '117@univ.edu'),
(118, 'Rhea Kapoor', 'Chemistry', 50, 8.4, '118@univ.edu'),
(119, 'Manish Agarwal', 'Mathematics', 95, 9.0, '119@univ.edu'),
(120, 'Tanvi Deshpande', 'History', 60, 8.1, '120@univ.edu');
```

Q 1: The academic office requires filtered student lists for preliminary analysis.

- Retrieve all students with **CGPA greater than 7.0**.
- Display all student records sorted by **Total_Credits in descending order**.
- List students from the **'Comp. Sci.'** department, ordered by **CGPA (highest to lowest)**.
- Display the **top 5 students** based on CGPA.
- Retrieve students whose **Total_Credits are less than the overall average credits**.

Q 2: Generate summary statistics for institutional reporting(Aggregate Functions).

- Find the **total number of students**.
- Compute the **average CGPA** of all students.
- Display the **minimum and maximum Total_Credits**.
- Count the number of students in each department.
- Calculate the **average CGPA for each department**.

Q 3: Perform department-level analytical evaluations(Grouping and HAVING Clause).

- Display departments with **more than three students**.
- Retrieve departments where the **average CGPA exceeds 7.5**.
- Identify departments whose **total accumulated credits exceed 200**.
- Display departments having **at least one student with CGPA below 6.0**.
- List departments ordered by **average CGPA (descending)**.

Q 4: Convert the following policy rules into SQL queries.

- Identify **high-performing students**
 - *(CGPA ≥ 8.5 AND Total_Credits ≥ 60)*
- Identify **at-risk students**
 - *(CGPA < 6.0 OR Total_Credits < 30)*
- List students **eligible for honors**, defined as:
 - CGPA ≥ 8.0
 - Total_Credits $\geq$ department average credits
- Retrieve students whose **CGPA is higher than their department's average CGPA**.
- Display departments where **no student has a CGPA below 6.5**.

Q 5: Solve the following using subqueries and/or HAVING clauses.

- Find the **second-highest CGPA** in the student registry.
- Retrieve students who share the **same CGPA with at least one other student**.
- Identify the department with the **highest average CGPA**.
- Display students whose **Total_Credits exceed those of all students in the 'History' department**.
- Generate a consolidated report containing:
 - Department Name
 - Total Number of Students
 - Average CGPA
 - Total Credits